Lect# 4:
Project Scope Management

-What is PSM?

- Strategic Planning
- Project Selection
- Scope definition & WBS
- Scope verification & scope Change Control

What is Project Scope Management?

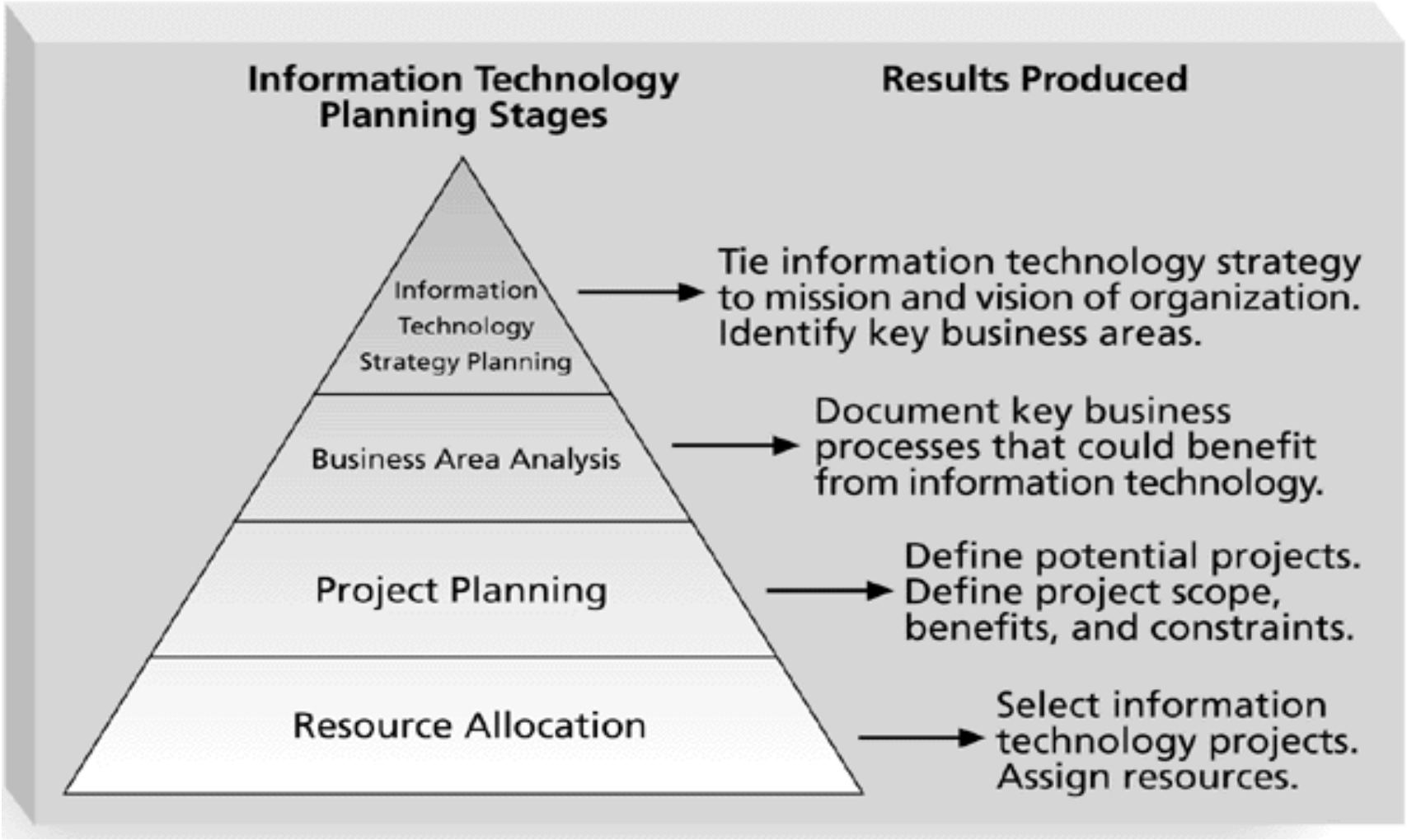
- Scope refers to all the work involved in <u>creating the products</u> of the project and the <u>processes</u> used to create them

- PSM Main Processes
  1. Initiation: beginning a project or continuing to the next phase
  2. Scope planning: developing documents to provide the basis for future project decisions
  3. Scope definition: subdividing the major project deliverables into smaller, more manageable components
  4. Scope verification: formalizing acceptance of the project scope
  5. Scope change control: controlling changes to project scope

1. Project Initiation: Strategic Planning and Project Selection

- The first step in initiating projects is to look at the big picture or <u>strategic plan</u> of an organization
  - Strategic planning involves determining long-term business objectives
  - IT projects should support strategic and financial business objectives

4 Stage Planning Process for Identifying IT Projects



Methods for Selecting Projects

- There are usually more projects than available time and resources to implement them
- Methods include
  - a: focusing on broad needs,
  - b: categorizing projects, and
  - c: financial methods

a. Focusing on Broad Organizational Needs

- It is often difficult to provide strong justification for many IT projects, but everyone agrees they have a high value
- Three important criteria for projects:
  - There is a *need* for the project
  - There are *funds* available
  - There's a strong *will* to make the project succeed

b. Categorizing IT Projects

- One categorization is whether the project addresses
  - a problem
  - an opportunity
  - a directive
- Another categorization is how long it will take to do and when it is needed
- Another is the overall priority of the project

c. Financial Analysis of Projects

- Financial considerations are often an important consideration in selecting projects
- Two primary methods for determining the projected financial value of projects:
  - Return on investment (ROI)
  - Payback analysis

Project Charters

- After deciding what project to work on, it is important to formalize projects
- A project charter is a document that formally recognizes the existence of a project and provides direction on the project's objectives and management
- Key project stakeholders should sign a project charter to acknowledge agreement on the need and intent of the project

Table 4-2. Sample Project Charter

- **Project Title:** Information Technology (IT) Upgrade Project
- **Project Start Date:** March 4, 1999 **Projected Finish Date:** December 4, 1999
- **Project Manager:** Kim Nguyen, 691-2784, knguyen@abc.com
- **Project Objectives:** Upgrade hardware and software for all employees (approximately 2,000) within 9 months based on new corporate standards. See attached sheet describing the new standards. Upgrades may affect servers and midrange computers as well as network hardware and software. Budgeted $1,000,000 for hardware and software costs and $500,000 for labor costs.
- **Approach:**
  - Update the IT inventory database to determine upgrade needs
  - Develop detailed cost estimate for project and report to CIO
  - Issue a request for quotes to obtain hardware and software
- Use internal staff as much as possible to do the planning, analysis, and installation

Table 4-2. Sample Project Charter (continued)

Roles and Responsibilities:

Name	Role	Responsibility
Walter Schmidt, CEO	Project Sponsor	Monitor project
Mike Zwack	CIO	Monitor project, provide staff
Kim Nguyen	Project Manager	Plan and execute project
Jeff Johnson	Director of IT Operations	Mentor Kim
Nancy Reynolds	VP, Human Resources	Provide staff, issue memo to all employees about project
Steve McCann	Director of Purchasing	Assist in purchasing hardware and software

Sign-off: (Signatures of all above stakeholders)

Comments: (Handwritten comments from above stakeholders, if applicable)
This project must be done within ten months at the absolute latest. *Mike Zwack, CIO*
We are assuming that adequate staff will be available and committed to supporting this project. Some work must be done after hours to avoid work disruptions, and overtime will be provided. *Jeff Johnson and Kim Nguyen, Information Technology Department*

2. Scope Planning and the Scope Statement

- A scope planning involves developing documents to provide the basic for future pjt decisions, include criteria for determining if the pjt or phase has been completed successfully
  - Inputs include pjt constraints, pjt assumption, pjt charter & descriptions of the products involved
- Output of this process is written scope statement

Scope Statement

- A scope statement is a document used to develop and confirm a common understanding of the project scope. It should include
  - a project justification
  - a brief description of the project's products
  - a summary of all project deliverables
  - a statement of what determines project success

3. Scope Definition and the Work Breakdown Structure(WBS)

- After completing scope planning, the next step is to further define the work by breaking it into manageable pieces
- Good scope definition
  - helps improve the accuracy of time, cost, and resource estimates
  - defines a baseline for performance measurement and project control
  - aids in communicating clear work responsibilities

Work Breakdown Structure(WBS)

- A WBS is an outcome-oriented analysis of the work involved in a project that defines the total scope of the project
- It is a foundation document in project management because it provides the basis for planning and managing project schedules, costs, and changes

Figure 4-6a. Sample Intranet WBS Organized by Product

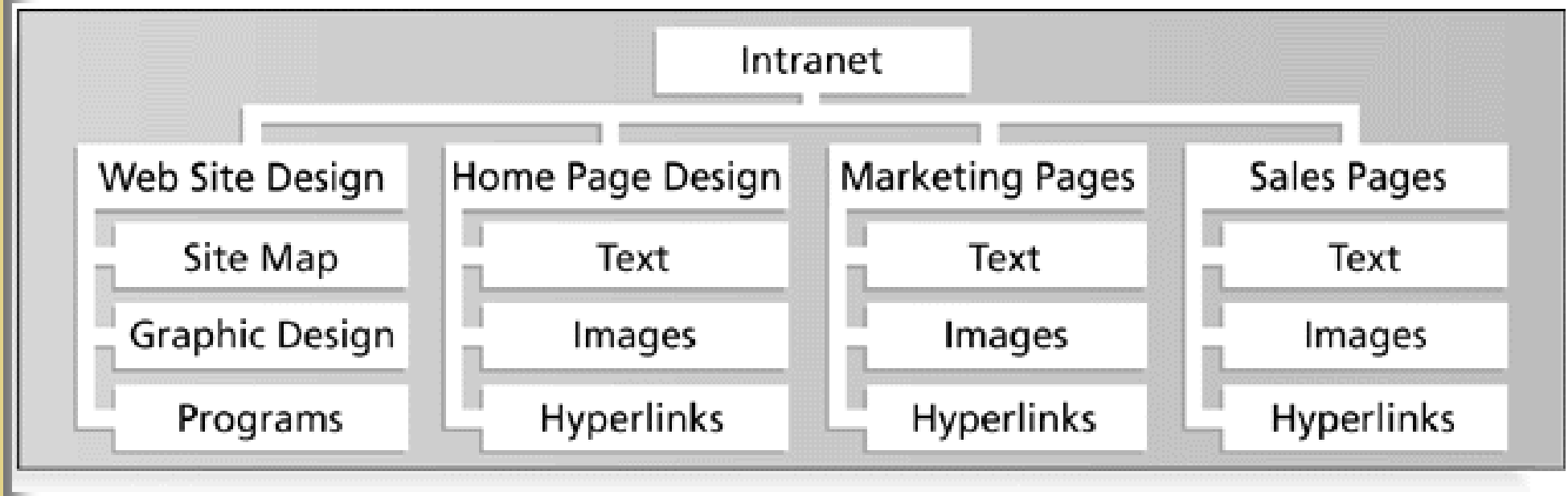


Figure 4-6b. Sample Intranet WBS Organized by Phase

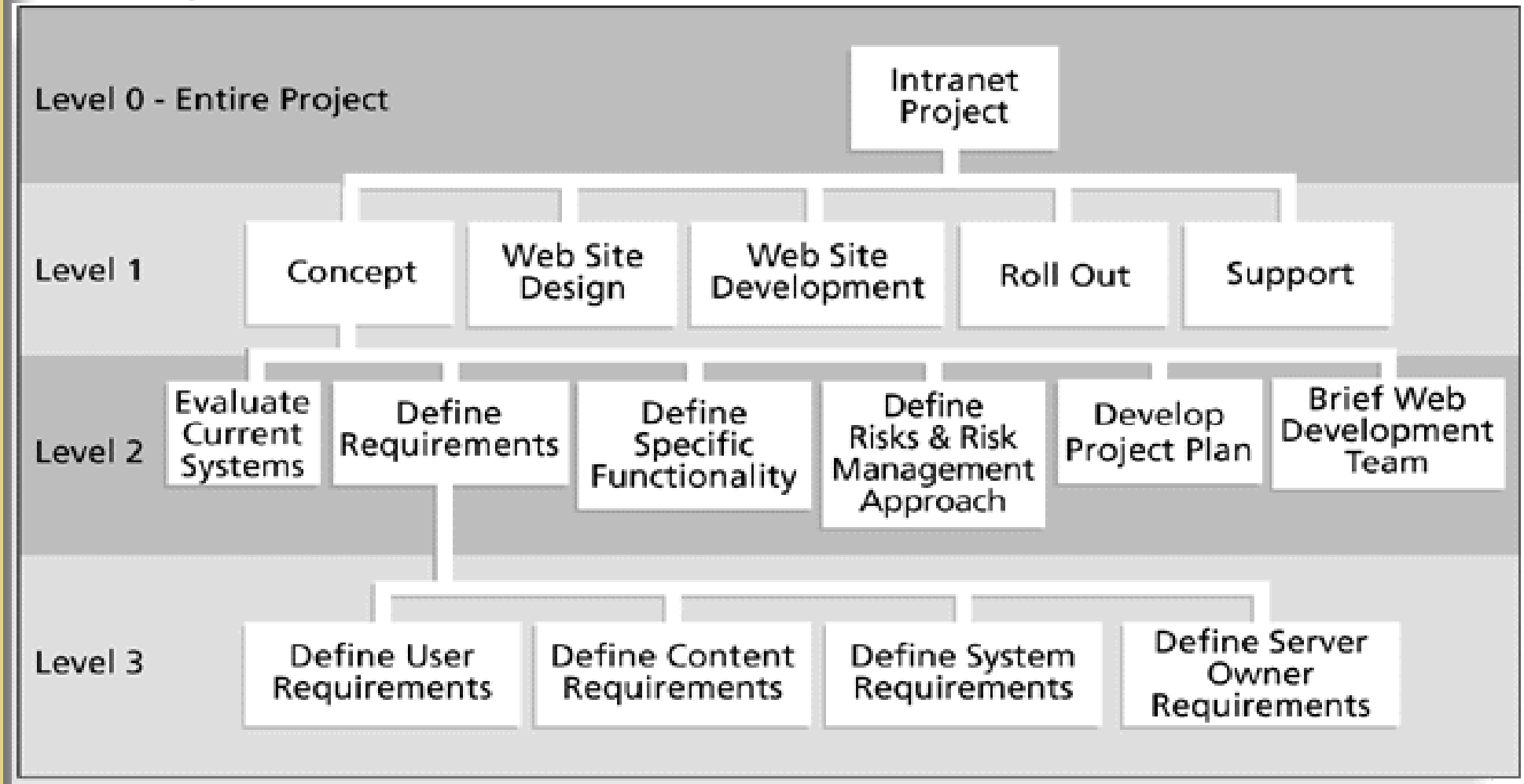
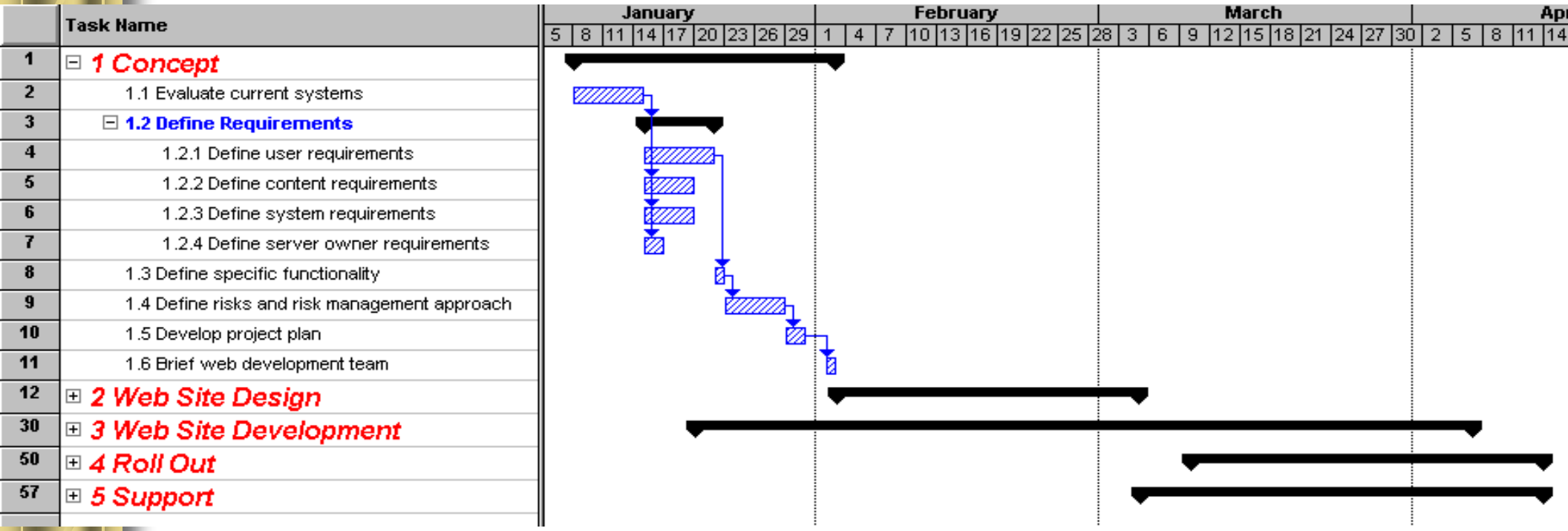


Table 4-3. Intranet WBS in Tabular Form

- 1.0 Concept
  - 1.1 Evaluate current systems
  - 1.2 Define Requirements
    - 1.2.1 Define user requirements
    - 1.2.2 Define content requirements
    - 1.2.3 Define system requirements
    - 1.2.4 Define server owner requirements
  - 1.3 Define specific functionality
  - 1.4 Define risks and risk management approach
  - 1.5 Develop project plan
  - 1.6 Brief web development team
- 2.0 Web Site Design
- 3.0 Web Site Development
- 4.0 Roll Out
- 5.0 Support

Figure 4-7. Intranet WBS and Gantt Chart in Project 98

WBS **Gantt Chart**



Project 98 file

Basic Principles for Creating WBSs

1. A unit of work should appear at only one place in the WBS.
2. The work content of a WBS item is the sum of the WBS items below it.
3. A WBS item is the responsibility of only one individual, even though many people may be working on it.
4. The WBS must be consistent with the way in which work is actually going to be performed; it should serve the project team first and other purposes only if practical.
5. Project team members should be involved in developing the WBS to ensure consistency and buy-in.
6. Each WBS item must be documented to ensure accurate understanding of the scope of work included and not included in that item.
7. The WBS must be a flexible tool to accommodate inevitable changes while properly maintaining control of the work content in the project according to the scope statement.

Cleland, David I. *Project Management: Strategic Design and Implementation*, 1994

4 & 5 : Scope Verification and Scope Change Control

- It is very difficult to create a good scope statement and WBS for a project
- It is even more difficult to verify project scope and minimize scope changes
- Many IT projects suffer from scope creep and poor scope verification
  - Telehealth
- Thus, very important to verify pjt scope & control pjt changes.

Table 4-4. Factors Causing IT Project Problems

Factor	Rank
Lack of user input	1
Incomplete requirements and specifications	2
Changing requirements and specifications	3
Lack of executive support	4
Technology incompetence	5
Lack of resources	6
Unrealistic expectations	7
Unclear objectives	8
Unrealistic time frames	9
New Technology	10

Suggestions for Improving User Input

- Insist that all projects have a sponsor from the user organization
- Have users on the project team
- Have regular meetings
- Deliver something to project users and sponsor on a regular basis
- Co-locate users with the developers
  - People know each other better by being close proximity

Suggestions for Reducing Incomplete and Changing Requirements

- Develop and follow a requirements management process
- Employ techniques such as prototyping, use case modeling, and Joint Application Design to thoroughly understand user requirements
- Put all requirements in writing and keep them current
- Create a requirements management database
- Provide adequate testing
- Use a process for reviewing requested changes from a systems perspective
- Emphasize completion dates

Exercise

1. **Create a WBS in chart form (same as Fig 4-6a) for the following projects:**
   - **Building your dream house**
   - **Create a new information system for your school**